# SpendSpace: Smart Expense Tracker and Personal Finance Management System

## Thesis Draft

Prepared for the degree requirements of Bachelor of Computer Applications / Bachelor of Technology / equivalent undergraduate program Project Category: Final Year Major Project Project Type: Full Stack Web Application Technology Stack: React.js, Node.js, Express.js, MongoDB-ready architecture, Local Fallback Data Store, Chart.js, Tailwind CSS

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## Title Page

**Project Title:** SpendSpace: Smart Expense Tracker and Personal Finance Management System **Submitted By:** _____ **Roll Number:** _____ **Department:** _____ **Institute:** _____ **University:** _____ **Session:** _____ **Submitted To:** _____

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## Certificate

This is to certify that the project entitled **"SpendSpace: Smart Expense Tracker and Personal Finance Management System"** submitted by _____ in partial fulfillment of the requirements for the award of the degree _____ is a bonafide work carried out under my guidance and supervision.

To the best of my knowledge, the work embodied in this project has not been submitted elsewhere for the award of any other degree, diploma, or certificate.

**Project Guide:** _____ **Head of Department:** _____ **External Examiner:** _____ **Date:** _____

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# Declaration

I hereby declare that the project report entitled **"SpendSpace: Smart Expense Tracker and Personal Finance Management System"** is my original work and has been prepared by me under the guidance of my supervisor. I further declare that this project has not been submitted to any other institution or university for the award of any degree or diploma.

**Name of Student:** _____ **Signature:** _____ **Date:** _____

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# Acknowledgement

I would like to express my sincere gratitude to my project guide, faculty members, and department for their valuable support, suggestions, and encouragement during the development of this project. Their academic guidance helped in shaping the technical and practical direction of the work. I am equally thankful to my friends and classmates for their support, feedback, and motivation throughout the project lifecycle.

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# Abstract

Modern users frequently struggle to maintain discipline in their personal finances because spending records are either distributed across notebooks, messaging applications, banking portals, or mental estimates that are rarely accurate. This problem becomes more serious for students, salaried employees, freelancers, and small families who need a simple, low-friction tool that helps them record, monitor, and analyze their day-to-day financial activity. The project **SpendSpace** is developed to solve this problem by offering a smart and visually polished web-based personal finance management system.

The system enables users to register and log in securely, create and manage expense as well as income entries, define monthly budgets, set savings targets, apply threshold-based alerts, and visualize financial behavior through dashboards, charts, summaries, and categorized transaction analysis. It also supports transaction filtering, export features, recurring item markers, wallet separation, payment method tracking, and insight-oriented presentation. The project is implemented using React.js for the frontend,

Node.js with Express.js for the backend, and a MongoDB-oriented data layer with a local fallback mechanism for development continuity.

The major contribution of this project lies in combining functionality with product-level user experience. Unlike very basic college projects that only perform CRUD operations, this system introduces an organized finance workspace with a cleaner interface, modular data design, better reporting, and professional presentation. The final outcome demonstrates how a full stack web application can be designed not only for technical completion but also for usability, maintainability, and real-world relevance.

**Keywords:** Expense Tracker, Personal Finance, Budget Management, Full Stack Web Development, React, Node.js, Data Visualization, Financial Analytics

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# Chapter 1: Introduction

Personal finance management is one of the most essential yet neglected digital habits in everyday life. A large number of people know approximately how much they earn, but very few maintain accurate and structured records of how they spend, save, or allocate their money across categories. This creates long-term problems such as budget leakage, low visibility of recurring expenses, inability to hit savings targets, poor decision-making, and difficulty in reviewing financial behavior.

In the current digital environment, several financial applications are available, but many of them are too complex, cluttered, heavily dependent on integrations, or unsuitable for small-scale personal use. Students and early professionals often need something much lighter: a system that is easy to use, visually clear, and capable of supporting day-to-day decisions without overwhelming the user.

This project introduces **SpendSpace**, a smart expense tracking and finance management system developed as a final-year academic project. The system is designed to provide a complete yet understandable environment for recording transactions, planning budgets, monitoring savings, and studying financial patterns through visual and textual insights.

The project follows a full stack model, where the frontend is responsible for the user experience and visual interaction, while the backend handles routing, persistence, validation, and data management. Special attention has been given not only to making the project technically functional, but also to making it look and behave like a practical software product.

The system includes modern features such as:

- User registration and login
- Expense and income entry management
- Budget and savings target configuration
- Alerts through threshold-aware dashboard indicators
- Category and payment method analytics
- Search, filter, and sorting options
- Export to Excel and PDF
- Recurring transaction indication
- Local data fallback when database service is unavailable

The inclusion of analytical and design-oriented elements makes this project much stronger than a traditional CRUD-only college application. It demonstrates understanding of frontend architecture, backend logic, user workflows, data shape evolution, UI polish, and deployment-ready thinking.

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# Chapter 2: Problem Statement

The central problem addressed by this project is the lack of an accessible, structured, and insightful system for managing personal financial records. In practical life, users face several common issues:

1. They do not consistently record expenses.
2. They cannot easily categorize their spending.
3. They cannot compare spending with budget goals.
4. They do not have a simple way to monitor savings progress.
5. They lack visual summaries that explain financial behavior.
6. Existing tools often prioritize complexity over clarity.

Many lightweight student projects solve only the basic part of the problem by allowing users to add or delete expenses. However, such systems do not meaningfully support planning, interpretation, or financial discipline. In real usage, people need more than a list of transactions. They need context. They need structure. They need answers to questions such as:

- Where is most of my money going?
- How much budget is left this month?
- Which expenses are still pending?
- How much income did I log against current spending?
- Am I progressing toward my savings target?
- Which category dominates my expenses?
- How have my finances changed across recent months?

Without a system that answers such questions, users continue making decisions with incomplete information. Therefore, the project problem is not merely data entry; the actual problem is financial visibility and control.

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# Chapter 3: Objectives of the Project

The project has been developed with the following primary objectives:

## 3.1 Main Objectives

- To build a web-based expense and income management system
- To allow users to maintain their own financial workspace securely

- To provide budgeting and savings planning features
- To improve visibility through data analytics and charts
- To offer a more professional user interface than a basic academic project

### 3.2 Functional Objectives

- To allow user registration and login
- To store user-level financial records separately
- To create, update, delete, and list transactions
- To distinguish between expense and income entries
- To include category, wallet, payment method, status, and notes in every record
- To set a budget, savings target, and alert threshold
- To filter transactions by type, category, date range, status, wallet, and method
- To export filtered records to PDF and Excel

### 3.3 Non-Functional Objectives

- To create a responsive and attractive interface
- To design the project with modular structure
- To keep the code maintainable and scalable
- To provide a fallback mechanism in development
- To prepare the project as a strong final-year submission

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## Chapter 4: Scope and Limitations

### 4.1 Scope of the Project

The project is intended for individual finance tracking. It can be used by students, salaried users, freelancers, and anyone who wants a private personal budgeting and expense analysis workspace. The system is suitable for:

- Monthly budgeting
- Daily spending recording
- Category-level analysis
- Personal savings planning
- Self-review before making better decisions

The project demonstrates the core structure of a personal finance application and can be extended into a larger SaaS product or mobile-first service in future work.

### 4.2 Limitations

Although the system is functionally strong for a final-year project, some limitations remain:

- No banking API integration is used
- No mobile application version is included
- No OTP/email verification is implemented
- No multi-user collaborative family account exists yet
- No cloud deployment-based database reliability is guaranteed in offline development mode
- Advanced machine learning-based prediction is not implemented yet

These limitations do not reduce the value of the project. Instead, they provide a realistic base for future expansion.

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## Chapter 5: Literature Review

The design of SpendSpace is informed by broader research and observations in personal finance systems, expense management tools, and dashboard-based decision support applications. Existing studies on digital finance behavior highlight that users are more likely to stay financially disciplined when their tools provide instant feedback, clean categorization, and visible trends rather than static record lists.

Many consumer finance applications use similar concepts:

- transaction logs
- budget allocation
- category grouping
- chart-driven analysis
- monthly review summaries

However, these systems are often built for mature commercial ecosystems. From an academic perspective, the challenge is to convert those real-world patterns into a practical and understandable project implementation. The gap between typical classroom CRUD projects and real-world products is large. This project attempts to reduce that gap by borrowing best practices such as:

- modular routes and models
- user-focused dashboard design
- form-driven workflows
- export-ready reporting
- analytics with meaningful business interpretation

The literature and industry trend both support the idea that visual and contextual finance tools improve user engagement. Therefore, this project was intentionally designed to move beyond plain record storage and toward insight-supported financial monitoring.

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# Chapter 6: Requirement Analysis

Requirement analysis was carried out by considering the needs of users who want a simple but complete personal finance tracker.

## 6.1 Functional Requirements

The system must:

- allow a new user to create an account
- allow an existing user to log in
- maintain separate records for each user
- allow adding expense and income entries
- allow editing and deleting existing entries
- allow the user to store title, amount, category, type, date, wallet, tags, notes, and payment method
- allow setting and updating a monthly budget
- allow defining a savings target
- calculate remaining budget and usage percentage
- generate category, payment, and trend analytics
- export filtered data to PDF and Excel

## 6.2 Non-Functional Requirements

The system should:

- be responsive on different screen sizes
- have a clean and professional interface

- load data quickly
- be readable and easy to use
- be simple to maintain and extend
- be robust enough to operate in development mode even when MongoDB is unavailable

### 6.3 Hardware and Software Requirements

**Hardware Requirements**

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 1 GB free space
- Display: Standard 1366x768 or above

**Software Requirements**

- Operating System: Windows 10/11, Linux, or macOS
- Node.js runtime
- NPM package manager
- Browser: Chrome, Edge, Firefox
- VS Code or equivalent editor

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## Chapter 7: System Design

The system design follows a layered full stack architecture.

### 7.1 High-Level Architecture

The architecture consists of three main layers:

1. **Presentation Layer**

Developed using React.js. Responsible for forms, navigation, dashboard rendering, charts, exports, and interaction workflows.

1. **Application Layer**

Developed using Node.js and Express.js. Responsible for routing, validation, request handling, and integration between frontend and storage.

1. **Data Layer**

Designed primarily for MongoDB using Mongoose schemas. For development resilience, a local JSON-backed fallback store is also included.

### 7.2 Data Flow

The user interacts with the frontend by submitting forms or pressing dashboard controls. The frontend sends HTTP requests to backend APIs. The backend processes requests, reads or writes data, and returns responses to the frontend. The frontend then updates UI state based on the response.

### 7.3 Major Design Considerations

- Keep the user workflow short and obvious
- Reduce friction in adding new transactions
- Keep important metrics visible above the fold
- Use charts only where they add understanding
- Ensure each module is understandable during viva or demonstration

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## Chapter 8: Tools and Technologies Used

### 8.1 React.js

React.js is used for creating the frontend interface. It supports component-based development and state-driven rendering, making it suitable for dashboards, forms, and modular UI blocks. React also makes it easier to update complex interfaces such as filters, analytics panels, and transaction modals.

### 8.2 Tailwind CSS

Tailwind CSS is used for styling. It enables utility-first design and speeds up UI implementation while retaining consistency. In this project, it helps achieve a cleaner and more professional visual appearance than traditional plain CSS approaches.

### 8.3 Node.js

Node.js is used as the server runtime environment. It allows JavaScript to be used on the backend and supports efficient development of RESTful APIs.

### 8.4 Express.js

Express.js is used to create backend routes for authentication, budget handling, and transaction management. It simplifies HTTP request handling and keeps the backend structured and understandable.

### 8.5 MongoDB and Mongoose

MongoDB is the intended database for persistent storage, while Mongoose provides schema-based interaction and model abstraction. The schema model supports structured user data, budget objects, and transaction fields.

### 8.6 Chart.js

Chart.js is used for data visualization. It enables category breakdowns, payment method analysis, and trend charts. The inclusion of charts improves the analytical strength of the application.

### 8.7 jsPDF and XLSX

These libraries are used to export records as PDF and Excel documents. The export functionality improves the academic and practical value of the project.

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## Chapter 9: Implementation Details

Implementation was carried out in iterative stages rather than a single-pass build. The early project stage supported only registration, login, and expense management. The final implementation expanded the application into a richer workspace.

### 9.1 Authentication Implementation

Authentication includes:

- registration form
- login form
- validation for email and password
- local user session storage
- redirect logic for protected routes

The system stores the current user in browser local storage so that the user session can survive refreshes. Protected and public routes are controlled in the React router layer.

## 9.2 Budget Management Implementation

The user can set:

- monthly budget
- savings target
- threshold alert percentage
- currency

These values are used throughout the dashboard to compute budget left, budget usage, and savings progress.

## 9.3 Transaction Management Implementation

Each transaction includes:

- title
- amount
- type
- category
- date
- payment method
- wallet
- status
- notes
- tags
- recurring flag

This richer structure significantly improves the quality of analysis and makes the project stand out academically.

## 9.4 Filtering and Search

The filter system supports multiple dimensions:

- search by keywords
- type selection
- category filtering
- date range filtering

- payment method filtering
- wallet filtering
- status filtering
- sorting options

This adds depth to the system and demonstrates practical frontend state handling.

### 9.5 Insight Generation

The dashboard provides meaningful text insights based on:

- budget threshold usage
- top spending category
- pending transactions
- savings target progress
- recurring entry presence

This is not full artificial intelligence in a strict academic sense, but it is smart rule-based interpretation, which strengthens the usefulness of the application.

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## Chapter 10: Module Description

### 10.1 Authentication Module

This module handles user registration and login. It validates basic credentials, prevents duplicate accounts, hashes passwords, and returns public user data after successful login.

### 10.2 Dashboard Module

The dashboard is the central working area of the system. It aggregates budget information, statistics, analytics, action buttons, filtering controls, and detailed transaction history.

### 10.3 Budget Module

This module allows users to define monthly financial boundaries. It acts as the planning unit of the application and connects directly to usage summaries and alert visualization.

### 10.4 Expense Module

This module handles the CRUD lifecycle of transactions. It supports both expenses and income and keeps the user's finance history dynamic and editable.

### 10.5 Reporting Module

The reporting module is responsible for exporting records in PDF and Excel formats. This feature is valuable during project demonstrations because it shows direct practical usefulness.

### 10.6 Visualization Module

This module produces:

- six-month trend charts
- category breakdown charts
- payment spread charts
- weekly pulse bars

These views improve the interpretive quality of the platform.

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## Chapter 11: Database and Data Handling Strategy

The project was designed with MongoDB as the primary persistence layer. Mongoose schemas were defined for:

- user
- transaction
- budget

The user model stores identifying information and authentication data. The transaction model stores the financial records associated with the user. The budget model stores configuration values for planning.

### 11.1 Need for Fallback Storage

During development, a common practical issue emerged: if MongoDB is unavailable locally, the backend cannot process database-driven routes, causing server-side failures. To improve project robustness during demonstration and testing, a local fallback data store was introduced.

The fallback strategy:

- detects when MongoDB is not connected

- writes and reads data from a JSON-backed local store
- keeps registration, login, budget, and transaction APIs functional

This is an important design strength because it allows the project to remain demo-ready even in the absence of local database services.

### 11.2 Data Integrity Considerations

Even in fallback mode, the project maintains:

- unique user identity
- transaction grouping by user
- budget grouping by user
- update timestamps
- structured object storage

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## Chapter 12: User Interface Design

One of the main strengths of this project is the attention given to UI and UX quality. Many academic projects are functionally correct but visually weak. This project attempts to improve that condition.

### 12.1 Design Goals

- modern but not overdesigned
- visually polished without harming usability
- responsive layout across common screens
- quick access to primary actions
- clearer information hierarchy

### 12.2 UI Features

- sticky navbar with section shortcuts
- polished dashboard hero
- clean action buttons
- well-grouped cards and panels
- professional modal form
- readable chart sections

- informative footer

### 12.3 UX Decisions

- important actions are placed near the top
- budget and overview are visible immediately
- filters are grouped in a dedicated workspace panel
- charts are shown only after context is established
- transaction details remain editable from the ledger

The design is not simply decorative. It helps users understand the application faster and complete tasks more comfortably.

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## Chapter 13: Testing and Validation

Testing was performed at multiple levels.

### 13.1 Functional Testing

The following functions were tested:

- registration
- login
- dashboard rendering
- adding transactions
- editing transactions
- deleting transactions
- setting budget values
- filtering transaction data
- export features

### 13.2 Build Verification

The frontend production build was executed successfully. This confirms that the React application compiles correctly and is ready for deployment or static serving in production mode.

### 13.3 Unit/Smoke Testing

Basic frontend tests were run and updated to align with the current application structure. Helper utilities were also validated.

### 13.4 Backend Verification

Backend files were syntax-checked, and authentication routes were tested directly using local HTTP requests. Registration and login were successfully validated after the fallback storage mechanism was introduced.

### 13.5 Validation Summary

The project was validated for:

- usability
- responsiveness
- route flow correctness
- data operation correctness
- export availability
- dashboard coherence

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## Chapter 14: Results and Discussion

The developed system achieved the major goals identified during requirement analysis. A comparison between the intended objectives and the final result shows strong alignment.

### 14.1 Achieved Outcomes

- A professional full stack personal finance application was developed
- User-specific financial workspaces were enabled
- Expense and income management were successfully integrated
- Budgeting and savings tracking were completed
- Analytics and insights were implemented
- Export and filtering functionality were added
- UI quality improved significantly over a conventional academic project

### 14.2 Academic Value

From an academic point of view, the project demonstrates the following competencies:

- full stack application design
- route and model organization
- frontend state management
- data visualization
- usability-oriented interface design
- issue diagnosis and fallback engineering

### 14.3 Practical Value

From a practical point of view, the project is actually usable. A student or working user can log transactions, review financial patterns, and perform monthly planning using the system.

### 14.4 Discussion

The project also highlights an important lesson in software engineering: robustness matters. A project that looks good but fails when a local database is unavailable is not reliable. By introducing a fallback store, the system became more demo-friendly and development-friendly. This also strengthened the real-world maturity of the project.

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## Chapter 15: Conclusion

SpendSpace successfully addresses the fundamental problem of scattered and poorly understood personal financial tracking. It provides a structured environment where users can record financial activity, define plans, and understand financial behavior through visual and contextual summaries.

The final implementation demonstrates that a student project can move beyond form submission and data listing into something much closer to a real product. By combining authentication, transaction operations, planning controls, charts, filters, exports, and refined interface design, the project achieves both functional and presentational strength.

The work also proves that careful iteration improves project quality. The application began as a standard expense tracker but was transformed into a richer finance workspace through design improvement, data model extension, analytics, testing, and resilience features.

In conclusion, the project is a strong example of how full stack web development can be used to solve a practical problem with academic depth and product-level clarity.

# Chapter 16: Future Enhancements

Although the current system is strong, several future enhancements can make it even better:

1. Cloud-hosted database and deployment pipeline
2. Email verification and password reset
3. Recurring automatic transaction scheduling
4. Multi-user family budgeting
5. Advanced AI-based spending recommendations
6. Monthly report emails
7. File attachments for bills and receipts
8. Mobile application version
9. OCR scanning for invoices
10. Goal-based financial forecasting

These enhancements can be treated as future research and development directions.

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# Chapter 17: Development Methodology

The development of SpendSpace followed an iterative and improvement-driven methodology rather than a rigid waterfall-only sequence. This was suitable because the project evolved from a basic expense tracker into a richer finance workspace with enhanced UI, analytics, and reliability.

## 17.1 Requirement Discovery

At the beginning, the major functions were identified:

- user onboarding
- financial record management
- budget planning
- dashboard summary
- export capability

As the project progressed, additional practical needs emerged such as recurring markers, filtering depth, wallet support, and local fallback storage. This confirms that iterative development is often more realistic for web projects than a completely fixed requirement model.

### 17.2 Incremental Build Strategy

The system was built in layers:

1. Base backend setup
2. Authentication routes and models
3. Expense CRUD operations
4. Budget routes and calculations
5. Dashboard integration
6. Data visualization
7. UI redesign and product polish
8. Stability and fallback improvements

This sequence reduced complexity because each layer became the foundation for the next.

### 17.3 Debugging and Refinement

The project required practical debugging beyond coding:

- package installation issues
- local environment startup issues
- missing MongoDB service
- route validation behavior
- frontend build compatibility

Handling such issues is an important part of real software development and strengthens the project academically.

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## Chapter 18: Feasibility Study

A feasibility study is important to understand whether the proposed system is realistic and sustainable.

### 18.1 Technical Feasibility

The project is technically feasible because:

- required tools are freely available
- JavaScript is used across frontend and backend
- the architecture is modular

- deployment paths are straightforward
- the system can run on ordinary student hardware

### 18.2 Economic Feasibility

The project is economically feasible because:

- no paid software is mandatory
- open-source libraries are used
- local development does not require expensive infrastructure
- hosting can be done on low-cost or free-tier services for demos

### 18.3 Operational Feasibility

The project is operationally feasible because the intended users do not require advanced technical skill to use it. The interface is form-driven and dashboard-oriented, which makes routine usage simple.

### 18.4 Schedule Feasibility

The project is feasible within an academic semester because it can be built and improved gradually. Core modules can be completed first, while design and reporting refinements can be layered later.

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## Chapter 19: Use Case Discussion

The system supports several practical user scenarios.

### 19.1 Registering a New User

The user opens the registration page, enters name, email, and password, and receives access to a personal dashboard.

### 19.2 Logging In

An existing user enters email and password. After validation, the user is redirected to the dashboard where the finance workspace is restored.

### 19.3 Adding a Transaction

The user opens the transaction modal and enters financial details such as title, amount, date, type, category, method, wallet, and tags. The entry is then saved and reflected in the dashboard.

### 19.4 Editing a Transaction

The user updates an existing record to correct an amount, note, or category. This supports data accuracy.

### 19.5 Budget Planning

The user sets a monthly budget and savings target. The system then calculates usage percentage and remaining balance.

### 19.6 Reviewing Analytics

The user studies visual charts to understand category spread, payment method usage, and long-term movement.

### 19.7 Exporting Reports

The user exports transactions in Excel or PDF format for review, record-keeping, or presentation purposes.

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## Chapter 20: DFD and UML Guidance

For final submission, the following diagrams should be created and inserted:

### 20.1 Context Diagram

Show a single system bubble called SpendSpace interacting with:

- user
- admin or guide (optional academic role)
- database

### 20.2 Level-1 DFD

Break the system into:

- authentication process
- transaction management process
- budget management process
- reporting process
- analytics process

## 20.3 Use Case Diagram

Actors:

- user

Use cases:

- register
- login
- add transaction
- update transaction
- delete transaction
- set budget
- view dashboard
- export data

## 20.4 Class Diagram

Suggested classes:

- User
- Expense
- Budget
- DashboardState
- AuthService

## 20.5 Sequence Diagram

Important sequence diagrams:

- user registration
- user login
- add transaction
- generate export

Including these diagrams will significantly increase the academic depth and final page count of the report.

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# Chapter 21: Security Considerations

Although the project is academic in nature, some basic security considerations were included:

- password hashing through bcrypt
- separation of public user data from password data
- per-user transaction scoping
- route-level request validation
- controlled frontend session storage

In production, the following additional security enhancements should be added:

- JWT or server-side session strategy
- secure cookies
- CSRF protection
- rate limiting
- stronger input sanitization
- HTTPS-only deployment

This chapter is useful in viva because it shows understanding beyond interface design.

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# Chapter 22: Testing Tables

## 22.1 Authentication Testing

| Test Case | Input | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| Register valid user | valid name, email, password | account created | account created | Pass |
| Register duplicate user | existing email | user already exists message | duplicate blocked | Pass |
| Login valid user | valid email and password | dashboard access | login successful | Pass |
| Login invalid password | wrong password | invalid credentials | invalid credentials shown | Pass |

### 22.2 Transaction Testing

| Test Case | Input | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| Add expense | title, amount, category, date | record saved | record saved | Pass |
| Edit transaction | changed amount | updated record | updated record | Pass |
| Delete transaction | valid ID | record removed | record removed | Pass |
| Filter by category | category chosen | matching list shown | matching list shown | Pass |

### 22.3 Budget Testing

| Test Case | Input | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| Set monthly budget | 50000 | budget saved | budget saved | Pass |
| Set savings target | 15000 | target saved | target saved | Pass |
| Trigger budget warning | high spend vs threshold | alert visible | alert visible | Pass |

### 22.4 Export Testing

| Test Case | Input | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| Export PDF | visible records | PDF generated | PDF generated | Pass |
| Export Excel | visible records | Excel file generated | Excel generated | Pass |

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# Chapter 23: Advantages of the Proposed System

The proposed system offers the following advantages:

- better than manual notebook-based expense tracking
- cleaner than cluttered spreadsheet workflows
- more insightful than simple transaction-only apps
- useful for students and professionals
- suitable for both academic evaluation and practical use
- strong frontend and backend integration demonstration

- modern design with real usability

The project therefore succeeds not only as a software artifact but also as an academic presentation asset.

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8. jsPDF Documentation. https://github.com/parallax/jsPDF
9. SheetJS Documentation. https://docs.sheetjs.com


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## Appendix A: Suggested Screenshots for Report

To expand the report into a rich final submission, include the following screenshots:

1. Registration page
2. Login page
3. Main dashboard overview
4. Budget planning panel
5. Filter workspace
6. Analytics charts
7. Ledger view
8. Transaction modal
9. Export workflow
10. Successful login and transaction addition

Each screenshot should be accompanied by a short explanation.

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## Appendix B: Suggested Viva Questions

1. Why did you choose React for the frontend?
2. Why is a fallback local store useful in this project?
3. How is user separation maintained in the system?
4. What is the difference between expense and income handling in your design?
5. How does your budget logic work?
6. What charts are used and why?
7. How can this project be scaled in future?
8. What makes this project better than a basic CRUD app?
9. What testing did you perform?
10. Which module was the most challenging to build and why?

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## Appendix C: Formatting Suggestion for Final 90-Page Submission

To convert this draft into a final academic report of approximately 90 pages:

- Use Times New Roman, 12 pt
- Use 1.5 line spacing
- Add page numbers
- Include institute certificate pages
- Add at least 10 to 15 screenshots
- Insert system design diagrams
- Insert use case diagrams, class diagrams, and flowcharts
- Expand testing tables and module screenshots
- Add implementation screenshots with explanation
- Add bibliography in required citation style

This thesis draft is intentionally structured so it can be expanded into a full final report with institution-specific formatting and screenshots.

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## Appendix D: Sample Screenshot Captions

1. **Figure 1:** Registration interface of the SpendSpace application
2. **Figure 2:** Login interface with secure access flow
3. **Figure 3:** Dashboard overview showing budget summary and key indicators
4. **Figure 4:** Filter workspace for category, date, and status-based review
5. **Figure 5:** Six-month financial trend visualization
6. **Figure 6:** Category-wise spending analysis
7. **Figure 7:** Payment method distribution chart
8. **Figure 8:** Detailed ledger with editable transactions
9. **Figure 9:** Transaction creation modal
10. **Figure 10:** Footer and final dashboard section

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## Appendix E: Sample Synopsis Paragraph

SpendSpace is a smart web-based personal finance management system developed to help users maintain organized records of income and expenses, set monthly budgets, monitor savings targets, and analyze spending patterns through interactive charts and dashboards. The system is implemented using React.js, Node.js, and Express.js with a MongoDB-ready data architecture and a local fallback store for development reliability. The project aims to improve financial visibility and user discipline through a professional interface and practical reporting tools such as PDF and Excel exports.

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## Appendix F: How to Increase the Thesis from Draft to Full Institutional Submission

If the department specifically asks for a strict 90-page report, the following additions should be included when formatting this file into Word:

- Title page, declaration, acknowledgement, certificate, and approval sheets
- Full page table of contents, list of tables, and list of figures
- Separate pages for each chapter introduction
- Large screenshots with explanation under each figure
- DFD, UML, ER, and flowchart diagrams
- Expanded module-wise source code explanation

- Step-by-step testing screenshots
- Deployment screenshots
- Future work diagrams and references in the required citation style

With proper formatting and screenshot inclusion, this draft becomes a strong base for a full 90-page academic report.